Nathan Janit

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Skills

Programming: Python, C++, C, JavaScript, Rust, php
Software: Omniverse (Kit, Isaac Sim, Create), Kubernetes, AutoDesk, SolidWorks, Power BI, Git
Libraries: TensorFlow, OpenCV, PyTorch, NumPy, MySQL, USD (Universal Scene Description)

Experience

Accenture – Eclipse Automation **Jan 2025 – April 2025**
Omniverse Testing Coordinator *Cambridge, Ontario*

- Debugged a web-based USD conversion pipeline, reducing error rates and downtime by 80%.
- Built and iterated on **Omniverse** workflows using Gaussian Splatting; extended viewport functionality with custom **Omniverse Kit** extensions.
- Created 3D animations of industrial machines using **Blender** and **SolidWorks**; integrated assets into Omniverse.
- Presented XR product visualizations using the **Apple Vision Pro** and **Magic Leap** headsets.
- Developed a dynamic layout planning extension using **Excel VBA** and **JavaScript** to automate workspace configuration within Omniverse.

Varicent **May 2024 – August 2024**
FP&A Intern *Toronto, Ontario*

- Created a computer vision-based bank reconciliation workflow using **Power Automate**, saving 5 hours each cycle.
- Designed a discrepancy discerning algorithm as an Excel macro to identify out-of-place entries in budget plans.
- Programmed an Excel workflow to web scrape the data from financial filings of the top public SaaS companies.

Silver Owl Robotics **May 2021 – Present**
Founder *Richmond Hill, Ontario*

- Founded the largest VRC program in Ontario, earning 40+ awards including **World Division Champion**.
- Built a company website and customer portal using **PHP**, managing registrations and payments.
- Automated operations via **Google Sheets** APIs for live budgeting, payroll, and employee scheduling.
- Taught CAD and robotics design using **Autodesk** and 3D printing workflows.

Projects

Real Time Automated Scoring | *Kubernetes, Rust, JavaScript, DeepStream* **March 2025 – Present**

- Developed a containerized web application that autonomously scores robotics competitions using real-time visual data.
- Reconstructed a virtual twin of the field via **multi-camera object tracking** using **DeepStream**.
- Implemented a probabilistic consensus filter to resolve detection inconsistencies and ensure accurate scoring.

Catan Multiplayer Game | *C++, Git* **November 2025 – December 2025**

- Developed a networked multiplayer version of *Catan* in base **C++**, deployed on a remote **Linux** server.
- Handled real-time input from multiple devices and implemented game logic using only the standard library.

Bot Detection- Datathon | *Python, MySQL, PyTorch* **February 2024 - March 2024**

- Developed an ML model with PyTorch to detect and flag non-human accounts based on transaction and network data.
- **Won the OwnersBox section of the 2024 CXC Hackathon for best technical implementation.**

UWAT VEXU | *C++, Autodesk, Git* **December 2023 - May 2024**

- Implemented closed loop and predictive correction algorithms to manage voltage control and reduce angle drift by 15%.
- Integrated a Pure Pursuit Algorithms which improved the accuracy of autonomous movement by 20%.
- **Ranked first out of over a hundred teams at the VEXU World Championship.**

Education

University of Waterloo **September 2023 – Current**
Bachelor of Computer Science *Waterloo, Ontario*
Wilfrid Laurier University **September 2023 – Current**
Bachelor of Business Administration *Waterloo, Ontario*